

PART I: Multiple Choice [24 pts]

Choose the best answer from the given alternatives and give your answer on the space provided in the answer sheet. [1 pt each]

1. Of the given alternative which one mismatched with its data unit in the case of TCP/IPv-2 reference model
 - A. Application Layer : Message
 - B. Transport Layer: Segment
 - C. Data-Link Layer: Ethernet Frame
 - D. Network Layer: Frame
 - E. All except D
2. Why is the User datagram protocol known to be connectionless, because
 - A. Every segment are treated independently by transport layer
 - B. It sends the pieces of information very quickly
 - C. It disseminate information in the way ordered
 - D. It passes the data as a stream of related segment
3. Assume a source address 2001:0db8:85a3::7225 and the destination ip address 10.45.58.120. Which IPv-6 reserved address is responsible for mapping the source address to the destination
 - A. Link-local Address
 - B. Compatible Address
 - C. Mapped Ip address
 - D. IPv-6 Converter
4. Identify a modulation scheme which utilize two different amplitudes of the carrier frequency to represent two binary level/values
 - A. Amplitude Modulation
 - B. Amplitude Shifting keying
 - C. Quadratic shifting Keying
 - D. Binary-amplitude shifting Keying
5. A multiple access protocol which don't allow station not to continuously sense, when the channel is busy, rather it waits a random amount of time and repeat the algorithm is
 - A. Persistent CSMA
 - B. Non-Persistent CSMA
 - C. Reservation
 - D. All
6. One explains synchronization service in session layer, is technique
 - A. which assign time for nodes during communication
 - B. Prevents a nodes for sending or receiving pieces of information simultaneously
 - C. That lets the nodes resume the communication after some sort of failure occurred
 - D. It prevent simultaneously critical operation
7. The OSI Reference Model provides for all of the following except
 - A. Defines the process for connecting two layers together, promoting interoperability between vendors
 - B. Allows vendors to compartmentalize their design efforts to fit a modular design, which eases implementations and simplifies troubleshooting
 - C. Separates a complex function into simpler components
 - D. Provides only theoretical help network administrators understand the communication process used between networking components

8. Discover an elements that are used to provide a reliable connection
- A. Ready/not ready signals
 - B. Windows
 - C. Sequence numbers and acknowledgments
 - D. Ready/not ready signals and windowing
9. Which of the following is not a feature of a protocol that is considered to match OSI Layer 4 ?
- A. Error recovery
 - B. Segmenting of application data
 - C. Flow control
 - D. Conversion from binary to ASCII
10. A type of bridge, which is responsible for converting data format in the data-link layer.
- A. Converter Bridge
 - B. Translational Bridge
 - C. Source Routing Bridge
 - D. Transparent Bridge
11. Which TDM variant where the output frame collects data from the input frame until it is full, not leaving an empty slot
- A. Synchronous TDM
 - B. Output TDM
 - C. Asynchronous TDM
 - D. All except B
12. What do we call a transmission impairment occur as a result of the effect of one wire on the adjacent wire
- A. Thermal noise
 - B. Crosstalk
 - C. Impulse noise
 - D. Induced noise
13. A typical type of cables which commonly used in Ethernet Local area Network:
- A. Copper Wire
 - B. Unshielded twisted-pair (UTP)
 - B. Shielded twisted-pair (STP)
 - D. Optical Fiber cable
14. Supposedly, you are testing connectivity with the ping command and getting a “request timeout” or “destination unreachable” message in reply. At what layer of the OSI Reference Model does this indicate a problem?
- A. Transport Layer
 - B. Data link Layer
 - C. Network Layer
 - D. Application Layer
15. Which of the following scenarios best demonstrates the application of the Dynamic Host Configuration Protocol (DHCP) in a network?
- A. A network administrator sets static IP addresses for each device in the company network manually.
 - B. A client device requests an IP address and other network configuration details from a DHCP server automatically when it connects to the network.
 - C. A router uses Network Address Translation (NAT) to allow multiple devices on a local network to share a single public IP address.
 - D. A switch uses the Spanning Tree Protocol (STP) to prevent network loops and ensure efficient data forwarding paths.
16. Which of the following protocols is responsible for ensuring that data packets are delivered in the correct sequence and without errors in a network?
- A. User Datagram Protocol (UDP)
 - B. Internet Protocol (IP)
 - C. Transmission Control Protocol (TCP)

D. Simple Mail Transfer Protocol (SMTP)

17. Which network service is used to resolve a hostname into an IP address?

A. File Transfer Protocol (FTP)

B. Domain Name System (DNS)

C. Dynamic Host Configuration Protocol (DHCP)

D. Hypertext Transfer Protocol (HTTP)

18. In the context of networking, what is the primary function of the Address Resolution Protocol (ARP)?

A. To determine the IP address of a given domain name.

B. To determine the hardware (MAC) address of a host from its IP address.

C. To allocate IP addresses dynamically to devices on a network.

D. To establish a secure channel between two networked devices.

19. Which protocol is primarily used to manage devices on an IP network, allowing administrators to monitor network performance, detect faults, and configure remote devices?

A. Simple Network Management Protocol (SNMP)

C. Internet Control Message Protocol (ICMP)

B. Hypertext Transfer Protocol (HTTP)

D. Post Office Protocol (POP)

20. Which of the following best describes the primary function of a router in a computer network?

A. A router encrypts data for secure transmission over the internet.

B. A router connects multiple devices within a local network to share files and printers.

C. A router forwards data packets between different networks, determining the best path for the data to travel.

D. A router translates domain names into IP addresses to route data to the correct destination.

21. What is the purpose of the routing table in a router?

A. To store a list of all the devices connected to the router.

B. To map IP addresses to MAC addresses within a local network.

C. To store information about the best paths to various network destinations.

D. To translate human-readable domain names to IP addresses.

22. At which OSI layer does a switch primarily operate, and what is its main function at that layer?

A. Layer 1 - Transmits raw bit streams over a physical medium

B. Layer 2 - Switches frames based on MAC addresses

C. Layer 3 - Routes packets based on IP addresses

D. Layer 4 - Ensures reliable data transfer

23. Which device operates at the OSI Network Layer and what is its primary function?

A. Hub - Amplifies and unicast data to all ports

B. Router - Forwards packets based on IP addresses

C. Switch - Forwards frames based on MAC addresses

D. Repeater - Regenerates signals to extend transmission distance

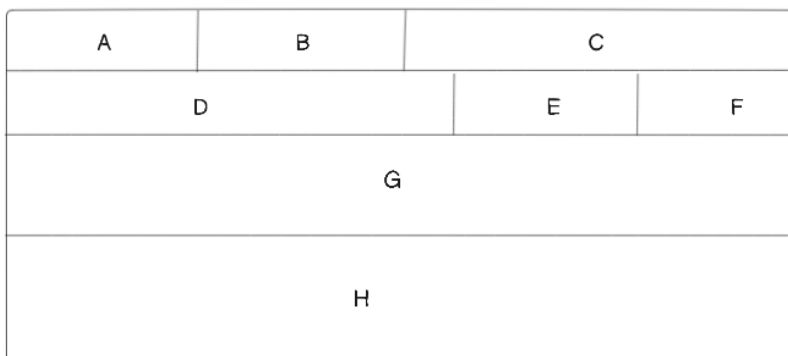
24. Which OSI layer is responsible for error detection and correction in data frames, and what device is typically associated with this function?

- A. Physical Layer - Repeater
B. Data Link Layer- Network Interface Card (NIC)
C. Transport Layer - Firewall
D. Application Layer)- Web Server

Part II: Short Answer [16pt]

Clearly and neatly answer the following the questions accordingly

1. Explain the key services in a session and presentation layer of OSI model. [3 pts]
2. Compare and contrast IPv-4 and IPv-6 in terms of size, representation and security. [3 pts]
3. Differentiate flow control and congestion control from the perspective of the transport layer. [3pts]
4. Explain the two types of routing algorithms. [2 pts]
5. The following diagram depicts the IPV6 datagram format. Explain the labeled field of the datagram briefly. (CLO-4) [5pt]



Part III: Encoding and Addressing [10pts]

Clearly and neatly answer the following the questions accordingly

1. Assume a PDU “GoodLuck” prepared at the Application layer. Apply Non-Return-to-Zero Inverted line encoding scheme to represent pieces of the PDU separately at the physical layer. [4 pts]
2. Suppose Ethio-Google enterprise has requested you to analyze the network topology below and design logical addressing. For future scalability the company gave you 172.20.12.0 as its default network with its default subnet mask. Based on the topology determine [6pts]

- Determine the Bit-to borrow for each NET
- The subnet mask of each NET
- The network address of each NET
- The address space of each NET
- The broadcast address of subnetwork

